

DATE	PARAMETER	EFFLUENT LIMITATION		REPORTED VALUES	
		Daily Maximum	Monthly Average	Daily Maximum	Monthly Average
5-19-15	TSS max 7-day	30 mg/L		51.50	
5-19-15	TSS max 7-day	800 lb/day		861.70	
5-19	CBODs max 7 day	270 lb/day		308.91	
5-19	CBODs max daily	10 mg/L		21.0	
5-19	Ammonia N	2 mg/L		2.46	
5-20	CBODs	10 mg/L		12.9	
5-20	Ammonia N	2 mg/L		3.08	
5-21	TSS	30 mg/L		68.00	
5-21	TSS	800 lb/day		1069.11	
5-21	CBODs max 7	270 lb/day		339.60	
5-21	CBODs	10 mg/L		21.60	
5-21	Ammonia	53 lb/day		77.35	
5-21	Ammonia	2 mg/L		4.92	
5-22	TSS	30 mg/L		87.00	
22	TSS	800 lb/day		1248.09	
22	CBODs	270 lb/day		309.86	
22	CBODs	10 mg/L		21.6	
22	Ammonia	53 lb/day		66.56	
22	Ammonia	2 mg/L		4.64	
5-25	TSS	30 mg/L		199.00	
5-25	TSS	800 lb/day		3337.03	
5-25	CBODs	270 lb/day		352.15	
5-25	CBODs	10 mg/L		21.0	
5-25	Ammonia	2 mg/L		2.87	
5-26	TSS	30 mg/L		60.00	
5-26	TSS	800 lb		972.82	
5-26	CBODs	270 lb		289.41	
5-26	CBODs	10 mg		17.9	
5-26	Ammonia	2 mg		2.28	
5-27	CBODs	10 mg/L		11.4	
5-28	CBODs	10 mg/L		12	
5-28	Ammonia	53 lb/day		55.76	
5-28	Ammonia	2 mg/L		3.40	
5-29	CBOD	270 lb/day		362.76	
5-29	CBOD	10 mg/L		21.00	
5-29	Ammon	53 lb/day		86.37	
5-29	Ammon	2 mg/L		5.0	
may	TSS max monthly	—	20 mg/L	—	28.7
may	CBODs max monthly	—	190 lb/day	—	190.47
may	CBODs	—	7 mg/L	—	11.87
may	Ammonia max monthly	—	24 lb/day	—	47.2
may	Ammonia max monthly	—	0.9 mg/L	—	3.0



Cadillac Utilities Department

WATER RESOURCES
DIVISION

FEB 29 2016

CADILLAC DISTRICT OFFICE

February 26, 2016

Mr. Jamie Wade, P.E.
Michigan Department of Environmental Quality
Cadillac District Office
120 West Chapin Street
Cadillac, Michigan 49601

Re: Violation Notice – Cadillac WWTP VN No. VN-006346

Dear Mr. Wade,

This is in response to your letter dated February 1, 2016 which you informed us of violations that occurred at the Cadillac waste water treatment plant. We will describe what happened and what was done to mitigate the situation. I believe that there were two separate incidents that took place leading up to the violations noted. The first period of violation took place from October 21, 2014 to November 27, 2014 and the second violation period was from January 9, 2015 to June 23, 2015.

At this time it isn't entirely clear what the influent or operational problems were because the plant supervisor that was in charge during the duration of violation periods is no longer here. I do know that there was an influx of material in both instances that caused a large mass of floating sludge on the primary clarifiers. This floating mass caused severe settling issues within the plant. The solids passing through the aeration basins into the secondary clarifiers' caused bulking within the clarifier tanks. The operators were instructed to continually increase the RAS and WAS, which increased the solids loading problem on the plant. The bulking made its way through the tertiary treatment causing our clothe filtration to bypass. The solids then made it through U.V. disinfection reducing U.V. dose and then into the river. I believe that a lack of communication and supervision led to the majority of these problems.

The increased unsettled solids passing through the plant caused the following violations on several occasions between October 21, 2014 and November 27, 2014:

- Total Suspended Solids 7-day Ave. mg/L

- Total Suspended Solids 7-day Ave. lbs./day
- CBOD5 lbs./day
- CBOD5 mg/L
- Fecal Coliform Max 7-day Gm.
- Ammonia as (N) mg/L
- Ammonia as (N) 7-day Ave. lbs./day
- Ammonia lbs./day
- Total Phosphorus

The following operational procedures were put into action to mitigate the issues dated above:

- Double pumping to reduce the solids in the primary clarifiers.
- Increased ferric chloride feed to aid in secondary settling.
- Increased aeration to provide more oxygen for the bacteria and to aid in driving off Ammonia.
- Mild Chlorine solution to control filamentous.
- Increased IPP monitoring to minimize the possibility of unwanted inflow.

The increased unsettled solids passing through the plant caused the following violations on several occasions between January 9, 2015 and June 23, 2015:

- Total Suspended Solids 7-day Ave. mg/L
- Total Suspended Solids 7-day Ave. lbs./day
- Total Suspended Solids Monthly Ave. Max
- Total Suspended Solids % Removal
- CBOD5 Monthly Ave. Max
- CBOD5 lbs./day
- CBOD5 mg/L
- Fecal Coliform Max 7-day Gm.
- Ammonia as (N) mg/L
- Ammonia as (N) 7-day Ave. lbs./day
- Ammonia Monthly Ave. Max
- pH

The following operational procedures were put into action to mitigate the issues dated above:

- Double pumping to reduce the solids in the primary clarifiers.
- Increased ferric chloride feed to aid in secondary settling.
- Increased aeration to provide more oxygen for the bacteria and to aid in driving off Ammonia.
- Mild Chlorine solution to control filamentous.
- Increased IPP monitoring to minimize the possibility of unwanted inflow.

At this time I would like to thank you for your time addressing the violations that took place. As I previously stated before the supervisor that was here at the time of the violation periods is no longer here. That being said I will never completely know all of the exact causes or events that took place. I also cannot give an explanation as to why the supervisor did not give written notice for each occurrence as is required in our NPDES permit. The only information I have is the vague violation reports I have received from the previous supervisor and information given to me by the operators. The poor communication and lack of supervision have been removed from the operations at the Cadillac waste water treatment plant. Through this I believe we will not have these operational issues again. Again I would like to thank you for your time.

Respectfully,

A handwritten signature in black ink, appearing to read 'Jeffrey M. Dietlin', with a long horizontal flourish extending to the right.

Jeffrey M. Dietlin
Director of Utilities
City of Cadillac
231-779-7346 direct line
231-775-0906 fax

State of Michigan Department of Environmental Quality

This is to certify that

Jeffrey M. Dietlin

having submitted acceptable evidence of qualifications, in accordance with the requirements of the Rules Governing the Certification of Sewage Treatment Works Operators, is qualified to operate any sewage treatment works of any of the following classifications:

A, B, C, D

Issued

March 4, 2014

Expires

April 15, 2017

Operator I.D. Number

10435

ISSUED BY THE DIRECTOR OF THE DEPARTMENT OF ENVIRONMENTAL QUALITY
UNDER AUTHORITY OF ACT 451, P. A. 1984, AS AMENDED

EQP 3402
REV 3/2014



DATE: 2-26-2016

TO: Michigan Department of Environmental Quality
Water Resources Division

ATTN: Brian Jankowski, Supervisor
Cadillac District Office

I hereby designate JEFFREY M. DIETRICH as the responsible operator in charge of the waste treatment facilities at CADILLAC W.W.T.P. discharge Permit No. MI0020257. He/she has met the requirements of the Department and has been properly certified. The Certificate No. is 10435, and the license classification(s) are A, B, C, D. The names, certificate number, and license classification(s) of other operators who are employed at the above facilities are listed below.

<u>Name</u>	<u>Certificate No.</u>	<u>License Classification(s)</u>
<u>Doug Langworthy</u>	<u>18398</u>	<u>C, D</u>
<u>DEBBIE GOODRICH</u>	<u>18456</u>	<u>C, D</u>
<u>KEVIN BRIGGS</u>	<u>19127</u>	<u>C, D</u>
<u>BRETT BALCOLM</u>	<u>19178</u>	<u>D</u>
<u>DAVE SKIFF</u>	<u>19190</u>	<u>D</u>

JEFFREY M. DIETRICH
Name
UTILITIES DIRECTOR
Title
CITY OF CADILLAC
Company

Send this form to: Brian Jankowski, Supervisor
Water Resources Division
Cadillac District Office
Department of Environmental Quality
120 West Chapin Street
Cadillac, MI 49601



Cadillac Utilities Department

December 21, 2015

Dear Mr. Jamie Wade,

On the afternoon of Friday December 18th an incident occurred at the Cadillac Waste Water Treatment Plant. An operator was performing a routine chemical cleaning of our Aqua Disk cloth filtration system. The operator had bypassed the three tanks and pumped the tanks solids to the head works of the plant. The operator then refilled the tanks with approximately 6,000 gallons each of tertiary water. Five gallons of Grease B Gone degreaser was added in total to the three tanks. Upon adding the degreaser the operator set the tanks to recirculate for a few hours. After recirculation the operator would normally pump the chemical solution to the head works where it can be treated before restoring the tanks to normal operation. Instead the operator returned the tanks to normal operation sending the chemical solution to our effluent. This took place around 12:30 p.m.

At approximately 1:30 p.m. fish in the onsite fish tank appeared to become incapacitated. The fish tank is feed with the plants effluent. At this time the operator was instructed to bypass the Aqua Disk filtration tanks stopping the chemical solution from going to our effluent. Immediately following this you were contacted and personnel were sent out to observe the effluent. Plant personnel walked the river banks looking for any sign of the degreaser in the river. There was no sign or indication the chemical had impacted the river in anyway.

Some of the fish in the fish tank and minnow net did expire but the majority of the incapacitated fish retuned to an active state within 20 to 30 minutes of redirecting the diluted cleaning solution. The Aqua Disk filtration tanks were then filled with tertiary water to further dilute and rinse the degreaser away. It was then pumped to the head works for treatment. This process was repeated three times before restoring the plant back to normal operation.

To prevent this or any other similar occurrences from happening in the future these processes will be gone over and more appropriate training will take place. Once again we would like to thank you for taking the time to come out and look things over.

A handwritten signature in black ink, appearing to read "Jeffrey M. Dietlin".

Jeffrey M. Dietlin
Utilities Director
City of Cadillac
(231)775-9326

WATER RESOURCES
DIVISION

DEC 21 2015

CADILLAC DISTRICT OFFICE

CITY OF CADILLAC MICHIGAN

200 North Lake Street • Cadillac, Michigan 49601
231.775.0181 • fax 231.775.8755
www.cadillac-mi.net

12/17/2015

Dear Mr. Jamie Wade,

As you know we are currently working on our Asset Management Plan for the City of Cadillac. We thank the MDEQ for the financial assistance opportunity in order to let us develop our detailed Asset Management Plan. We were awarded the grant in Round 2 and we plan to have the full Asset Management Plan completed before the end of the SAW Grant period. We understand that in order to satisfy the current permit requirements that you need the schedule for our Asset Management Plan (AMP) work.

Our AMP schedule is:

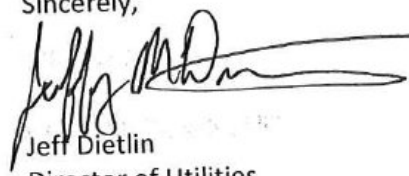
Current Work

Perform data collection and Assessing Condition
Perform Level of Service and
Risk and Consequence of Failure Analysis
Write Asset Management Plan
Complete Asset Management Plan
Determine Rate Structure/User Charges

Spring/Summer 2016
Fall 2016 (9/30/16)
Winter/Spring 2017

We have attached Section 5.0 of our Operation and Maintenance Manual addressing Personnel Requirements as well as the Organization Chart for the Utilities Department for the City which is consistent with the change in the NPDES permit. If you would like, we can also provide you with our current system map that of course will be updated during our SAW Grant work. Otherwise we will wait on a map submittal until it has been updated with the field collected data. Please let me know if you require any additional information for reissuance of our NPDES permit.

Sincerely,



Jeff Dietlin
Director of Utilities
City of Cadillac
231-775-0181 ext 124

WATER RESOURCES
DIVISION

DEC 21 2015

CADILLAC DISTRICT OFFICE

Stantec
CITY OF CADILLAC
WASTEWATER TREATMENT PLANT
OPERATION AND MAINTENANCE MANUAL

5.0 PERSONNEL

5.1 PERSONNEL REQUIREMENTS

The operation of the Cadillac WWTP and collection system is accomplished by qualified personnel. For routine operation of the WWTP, it is expected that personnel will be required eight hours per day, five days a week, and one operator per day on call during weekends and holidays, in addition to laboratory requirements. The collection system, sewers and forcemains is maintained by the City's Utilities Department. However, the collection system lift stations are maintained by the WWTP staff. The organizational structure of the Utilities Department is outlined in **Appendix F**.

5.2 CERTIFICATION

The Department of Environmental Quality, Municipal Wastewater Division, General Rules, requires that the wastewater plant operator (the person signing the monthly operating report) be certified, and that the certification be equivalent or higher than the classification of the treatment plant.

The supervisor of the Cadillac WWTP plant is required to have a Class "B" license, according to the general rules.

WATER RESOURCES
DIVISION

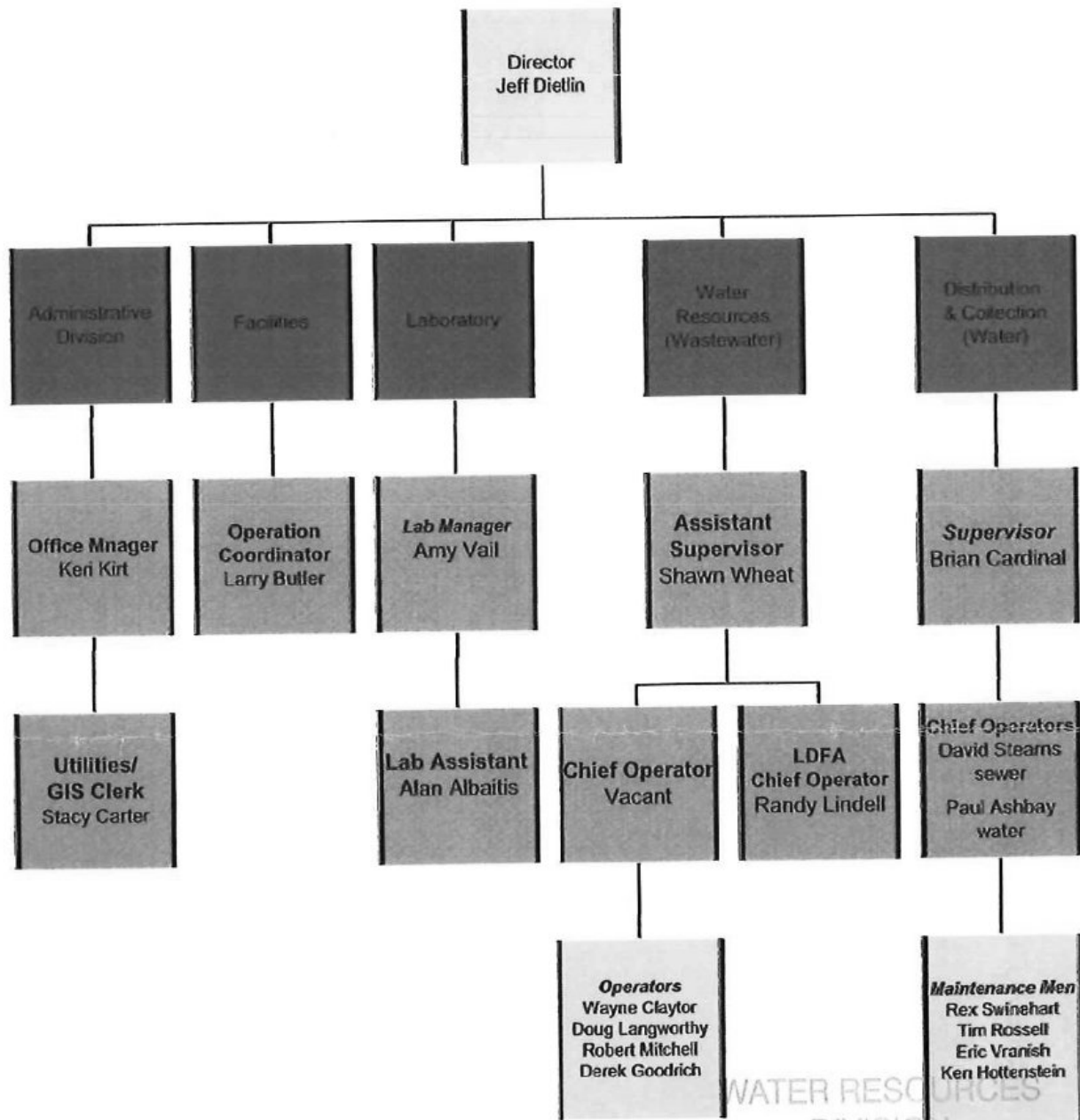
DEC 21 2015

CADILLAC DISTRICT OFFICE

♦ Original O & M Manual prepared by ALNM 1975, was used in the preparation of this revision.

**CITY OF
CADILLAC
MICHIGAN**

Utilities Department



WATER RESOURCES
DIVISION

DEC 21 2015

CADILLAC DISTRICT OFFICE

Note to the File

To: Julie Lowe
From: Julie Lowe
Date: October 16, 2015
Re: Michigan Rubber

Hamid left the Cadillac WWTP and replacing him for the moment is Jeff Dietlin. Jeff mentioned that issues with oil and grease at the facility have been resolved. Jeff contacted Michigan Rubber and dischargers to the Haring Township line and told them that someone is discharging oil and grease and if it continues, the city will fine the facilities found responsible. Michigan Rubber's contribution was less than half of what was coming from the Haring Township lift station.

As of October 2015, this issue is resolved.

I spoke to Jeff about completing a categorical determination for Michigan Rubber as mentioned in Hamid's letter dated July 8, 2015. Jeff believes they are categorical and not sure why they were listed only as an SIU. The Cadillac WWTP samples monthly at the facility and are treating them as if they are categorical. This will change in the next annual report.

CITY OF CADILLAC MICHIGAN

200 North Lake Street • Cadillac, Michigan 49601
231.775.0181 • fax 231.775.8755
www.cadillac-mi.net

Received
WILLIAMS

JUL 14 2015

GAYLORD FIELD OFFICE

July 08, 2015

Julie Lowe
Water Resources Division
Gaylord Field office
2100 W. M-32
Gaylord, MI 49735-9282

Dear Ms. Lowe,

Subject: Pretreatment Compliance Inspection
NPDES Permit MI0020257
Designated Name: Cadillac WWTP

This is a response to your letter dated June 26, 2015 with the subject above attached herein.

Item 1.

I have done the annual IPP for the Cadillac WWTP since 2011 and I did not know that I had to publish all Categorical Industrial Users in SNC in a local paper. We have a good working relationship with the local industries in town and publishing this information will tarnish our relationship. We have been visiting and communicating with our permitted industries throughout the year and try to resolve any issues in regards to their operation and their discharge limits.

Item 2.

We are not sure if Michigan Rubber was responsible for what was affecting the WWTP. We did Fats, Oils, and Grease (FOG) as to see the possible source of grease coming to the plant from an Industrial User. Michigan Rubber had a detectable hit on FOG. The FOG concentration through Haring TWP main sewer line was also analyzed and was much greater than any other FOG readings. Since June 16, 2015 the Haring WWTP has been receiving that flow and it is no longer a part of our collection system.

We lost our nitrification ability toward the end of January 2015. As you know, growing nitrifying bacteria in the winter is difficult especially with one of the coldest winters we have had on the record. We also had problems with filamentous bacteria that cause poor settling and bulking. A toxicity issue was a concern, and so we did various tests on the incoming flows to the plant. It is also possible that we were getting flows from a non-permitted or a non-industrial location.

I visited and spoke to various permitted facilities about their discharge and their permits and had several walk-throughs. Currently the WWTP is meeting the ammonia limits and is running efficiently with 96-98% TSS removal efficiency.